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RESEARCH ARTICLE

CHRExpert: An AI-Driven Court of Human Rights Expert Assistant for Legal Practitioners Utilizing Transformer Models

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ABSTRACT The growing complexity of human rights litigation, combined with the increasing volume of legal documents, presents significant challenges for legal professionals in efficiently managing case preparation and decision-making. This paper introduces CHRExpert, an Artificial Intelligence (AI)-driven legal assistant designed to support law practitioners in analyzing judicial decisions and predicting case outcomes. Utilizing a Pretrained Generative Transformer (GPT) with 6 billion parameters, fine-tuned on the European Court of Human Rights (ECHR) dataset, CHRExpert automates legal document analysis by identifying key legal provisions, performing statutory interpretation, and applying analogical reasoning to suggest legal strategies. The system is designed to support practitioners in aligning their legal arguments with judicial reasoning, particularly in post-litigation analysis. Evaluations based on final judgments demonstrated an accuracy of 83% in predicting outcomes for cases involving Articles 3, 6, and 8 of the European Convention on Human Rights, with an average AUC of 0.93. Additionally, CHRExpert reduced the time required for case preparation by 40%, streamlining legal workflows and improving efficiency. This paper demonstrates the potential practical application of AI to enhance legal processes by offering accurate predictions and experiential insights, making it a valuable tool for human rights litigation.

INDEX TERMS AI legal assistant, human rights litigation, transformer models, GPT-J, case outcome prediction, legal document analysis.

I. INTRODUCTION

The complexity of human rights litigation and the large number of legal documents involved in judicial processes make it challenging for legal professionals to manage case preparation and decision-making efficiently [1]. Courts and legal practitioners are required to review this large number of case laws. It is a process of interpreting statutes and analyzing legal precedents to form arguments and predict outcomes [2]. In recent years, there has been growing interest in the use of Artificial Intelligence (AI) to support legal processes. It demonstrates the impressive potential to analyze legal documents and provide insights based on past cases [3]. However, existing tools often struggle to capture the complex context of human rights cases and the role of judicial discretion [4]. This has highlighted the need for a system capable of not only automating legal document analysis but also offering accurate predictions and recommendations that align with judicial reasoning. The proposed system, CHRExpert, addresses this gap by leveraging advanced transformer models to assist legal professionals in handling complex human rights litigation.

The CHRExpert has been trained on the comprehensive dataset on human rights recognized as the European Court of Human Rights (ECHR) dataset [5]. It is a rich dataset with enough judicial decisions and associated legal provisions to train a transformer model. The CHRExpert has been developed using a Generative Pretrained Transformer (GPT) model, a 28-layer Deep Neural Network (DNN) with 6 billion trainable parameters [6]. Each layer of the CHRExpert is a transformer layer consists a token embedding layer, positioning encoding layer, multi-head attention layer,

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feed-forward network, and normalization layer with residual connection [7]. It leverages the potential of transfer learning [8] to utilize the existing language analysis capability of the GPT and extend it to interpret statutory language, spot key legal issues, and recommend legal strategies based on past precedents. The system also includes functionalities for analogical reasoning, which helps law practitioners draw parallels between current and past cases, further assisting in legal strategy development. CHRExpert's deployment includes an easy-to-use user interface, providing legal professionals with a streamlined experience to upload documents, view predictions, and access relevant case information. The core contributions of this paper are as follows:

- Development of CHRExpert, an AI-driven legal assistant, capable of analyzing human rights case documents and predicting case outcomes based on statutory interpretation and legal precedents.
- Extending the capability of the GPT model to interpret complex legal texts, identify key issues, and apply analogical reasoning.
- Evaluation of CHRExpert's performance, achieving an accuracy of 83% in case outcome prediction with an AUC of 0.93, alongside a 40% reduction in case preparation time for legal professionals.
- Developing a practical AI-integrated solution that handles legal document submission, predictive analysis, and legal strategy recommendations to streamline litigation workflows.

It is important to note that CHRExpert is designed as a post-litigation legal analysis assistant, aiding practitioners in understanding judicial reasoning and formulating legal strategies based on precedents and final rulings. It is not intended as a predictive tool for assessing applications before a case is heard. This distinction ensures that CHRExpert aligns closely with the realities of legal practice, where practitioners often work with judgments to shape their litigation strategies.

This paper aims to address the following key research questions:

- **RQ1:** How can transformer-based AI models be effectively applied to analyze human rights case documents and predict case outcomes?
- **RQ2:** To what extent can an AI-driven legal assistant improve the efficiency of case preparation and legal strategy development for human rights litigation?
- **RQ3:** What level of predictive accuracy and performance can be achieved by a transformer model trained on ECHR judicial data for case outcome prediction and legal document analysis?

These research questions guide the development, evaluation, and discussion of CHRExpert throughout this study.

The remainder of the paper is structured as follows: Section II provides a review of the existing literature on AI in legal applications and highlights the challenges in current systems. Section III explains the methodology, including the dataset, preprocessing steps, and the implementation of the transformer model. The implementation process of the proposed CHRExpert is presented in Section IV. After that, Section V presents the experimental analysis, detailing the system's performance metrics such as accuracy, AUC, and time efficiency. Section VI discusses the limitations of the proposed system, followed by future directions for improving CHRExpert's capabilities. Finally, Section VII concludes the paper by summarizing the findings and outlining potential advancements for the system.

II. LITERATURE REVIEW

Artificial intelligence (AI) 's rapid development has significantly influenced many fields, including law [9]. In recent years, the rise of conversational AI powered by transformer-based models has demonstrated great potential in legal applications such as document review, legal research, and outcome predictive analytics [10]. This literature review provides an overview of previous research on using AI in the legal domain.

Brożek et al. [11] explored the potential of Machine Learning (ML) in legal decision-making from real and imaginary situations. Barysė and Sarel [12] explored if AI is effective in judicial decision-making. The research trends discovered by Sil et al. [13] suggest that AI's application in legal decision-making has gained popularity because of demonstrating the potential support in decision-making for legal professionals. However, most state-of-the-art AI applications suffer from flexibility issues, and their ability to handle unstructured legal text is a significant research gap in this field. Widely used ML models, such as Support Vector Machines (SVM) [14], Random Forests (RF) [15], and K-Neighbour Classification (KNN) [16], showed some level of success. However, they are constrained by their inability to capture the context-dependent nature of legal language. The proposed CHRExpert is a transformer-based model capable of learning from unstructured data and capturing the complex context-dependent characteristics of legal aspects.

According to Maley [17], legal texts are inherently complex because of dense language. Specific terminology references to precedents and statutory laws and the contextual relation of these with the case description complicates legal language [18]. Besides, different types of interpretation of the same scenario based on the emphasis on different aspects make it more complicated [19]. That is why transformer models are capable of making analogical reasoning and balancing competing principles appropriate for legal decision-making. The introduction of the transformer model by Vaswani et al. [20] in 2017 marked a paradigm shift in Natural Language Processing (NLP). Unlike Recurrent Neural Networks (RNNs) [21] and Long Short-Term Memory (LSTM) networks [22], the transformer model is capable of processing text in parallel and capturing long dependencies within text. As a result, it is suitable to understand complex legal documents.

III. METHODOLOGY

A. DATASET AND PREPROCESSING

Dataset and its preprocessing play a crucial role in the overall effectiveness of the Deep Learning approach [23]. The proposed CHRExpert is an AI-driven assistant that helps law practitioners with human rights issues. Such a system needs to learn the complex features of judicial documents. The dataset processing method used in this section ensures that it is processed correctly and suitable for the transformer model to learn from. The entire process has been expressed in Algorithm 1, and later, the conceptual background has been explained.

Algorithm 1 Preprocessing Steps for CHRExpert Model Training

Input: ECHR dataset $D = \{d_1, d_2, \ldots, d_{11000}\}$
Output: Preprocessed dataset $D' = (X, Y)$
for each document $d_i \in D$ **do**
  Clean and Normalize: $T(d_i) \leftarrow L(d_i), S(d_i), A(d_i)$
  Filter Outcome-Dependent Terms: $T'(d_i) \leftarrow F(T(d_i))$
  Tokenize: $\tau(T(d_i)) \leftarrow$ BPE Tokenization
  Embed Tokens: $E(t_i) \leftarrow \mathbf{M} \times \text{OneHot}(t_i)$
  Pad/Truncate: $S' \leftarrow$ adjust sequence to length L
  Generate Mask: $M \leftarrow$ create mask for S'
  Form Pairs: $(X, Y) \leftarrow (S', \text{Outcome})$
end for
Split Dataset: $D' \leftarrow (D_{\text{train}}, D_{\text{val}}, D_{\text{test}})$ with ratio 70:15:15

The filtering function $F(T)$ is designed to remove outcome-dependent terms such as "damages," "pecuniary," "compensation," and other indicators that typically appear only when a violation is found [24]. This prevents the model from relying on outcome-revealing sections and ensures that CHRExpert predicts outcomes based solely on the facts and legal reasoning present before judgment determination.

1) EXPLORATORY ANALYSIS

Medvedeva et al. [25] analyzed 584 cases and corresponding documents publicly available in the European Court of Human Rights (ECHR) database and produced the "Judicial Decisions of the European Court of Human Rights: Looking into the Crystal Ball" dataset. It is known widely known as the ECHR dataset. It is an ideal dataset for training transformer models due to its rich and complex legal texts comprising 11,000 files. Each of these files contains a judicial decision and detailed information on the facts, legal arguments, and outcomes of cases heard by the ECHR. The dataset exclusively comprises final judgments issued by the European Court of Human Rights (ECHR). Each document reflects the court's conclusive ruling, including detailed judicial reasoning, legal provisions, and factual evaluations. Applications, communications, or procedural filings were not part of the dataset, ensuring that CHRExpert was trained on authoritative legal decisions. The elements of this dataset structure are listed in Table 1.

Furthermore, we analyzed the distribution of cases across different jurisdictions in the ECHR dataset. This analysis revealed that certain countries, such as Ukraine, Russia, and Turkey, had a disproportionately higher number of cases. This is a known characteristic of ECHR case data, reflecting regional variations in human rights adjudication. Table 2 provides an overview of the country-level distribution in the dataset.

2) TEXT CLEANING AND NORMALIZATION

The dataset preprocessing for the proposed CHRExpert starts with text cleaning and normalization. It ensures that the input data are consistent, noise-free, and in a standardized format [26]. As a result, the transformer model learns more effectively. Equation 1 shows the text cleaning and normalization process where $L(x_i)$, $S(x_i)$, and $A(x_i)$ represents lowercasing, stop-word removal, and alpha-numeric character removal functions, respectively [27].

$$T(x) = \{\mathrm{L}(x_i) \mid x_i \in x \wedge \neg\mathrm{S}(x_i) \wedge \mathrm{A}(x_i)\} \tag{1}$$

Furthermore, to mitigate data leakage arising from outcome-revealing terms, CHRExpert's preprocessing pipeline incorporates a filtering step to remove sections containing terms such as "damages," "pecuniary," and "compensation." These terms commonly appear only when a court rules in favor of the applicant, and their presence could artificially inflate prediction accuracy [28]. By filtering these terms during preprocessing, CHRExpert ensures that outcome predictions are grounded in legal reasoning and case facts, rather than post-judgment monetary awards. After text cleaning and normalization, the dataset is converted into all lowercase characters without any stop-word or alphanumeric characters.

3) TOKENIZATION

The transformer model used in the development of CHRExpert accepts tokens directly instead of plain text. That is why the cleaned and normalized texts have been transformed into tokens expressed in Equation 2. The Byte-Pair Encoding (BPE) [29] method has been used to manage the vocabulary size and improve model generalization while forming the tokens.

$$\tau(x) = \{t_1, t_2, \ldots, t_n\} \tag{2}$$

In Equation 8, $\tau(x)$ is the tokenization process for a sentence x where t_i represents individual tokens resulting from the tokenization process.

4) INPUT EMBEDDINGS

The input layer of the transformer model utilized to develop the proposed CHRExpert requires a dense vector as input. The tokens prepared using Equation 2 are further processed using Equation 3 to embed into dense vectors $E(t_i)$ is the

TABLE 1. The dataset structure analysis.

<table>
<tr><th>Case Metadata</th><th>Case Facts</th><th>Legal Provisions</th><th>Judgment Information</th><th>Additional Features</th><th>Additional</th></tr>
<tr><td>Application Number</td><td>Fact Summary</td><td>Articles Invoked</td><td>Outcome</td><td>Case Themes</td><td>Annotations</td></tr>
<tr><td>Applicant and Respondent</td><td rowspan="2">Legal Arguments</td><td rowspan="2">Legal Keywords</td><td>Judgment Text</td><td rowspan="2">Country-Specific Information</td><td rowspan="2">Cross-References</td></tr>
<tr><td>Decision and Application Date</td><td>Dissenting Opinions</td></tr>
</table>

TABLE 2. Country-wise distribution of cases in ECHR dataset.

Country	Number of Cases
Ukraine	2800
Russia	2200
Turkey	1500
Romania	1200
Italy	800
Other Countries	3500
Total	**11000**

embedding data, which captures the semantic meaning [30].

$$E(t_i) = \mathrm{M}(V \times d) \times \mathrm{OneHot}(t_i) \tag{3}$$

In Equation 3, $\mathrm{M}(V \times d)$ is the embedding matrix with $V \times d$ dimension where V is the vocabulary and d is the dimension. The embedding matrix is multiplied with the one-hot encoded vector for token t_i.

5) SEQUENCE PADDING AND TRUNCATION

Having uniform length (L) for all input sequences is essential to avoid training errors. Sometimes, the input sequence length is shorter, and sometimes, they are longer than the vector dimension of the input layer. For shorter sequences, padding is used. The transition is applied to longer sequences. For an input sequence $S = \{t_1, t_2, \ldots, t_n\}$ of length n, the conditional relation expressed in Equation 4 applies padding or truncation according to the sequence length [31].

$$S'_i = \begin{cases} t_i & \text{if } i \leq n \text{ and } i \leq L, \\ [\mathrm{PAD}] & \text{if } i > n \text{ and } i \leq L, \\ 0 & \text{if } i > L. \end{cases} \tag{4}$$

In Equation 4, S' is the padded or truncated sequence of fixed length L.

6) ATTENTION MASKING

The padding performed in the previous section raises unexpected attention to meaningless data in the input sequence. The attention masking expressed in Equation 5 has been used to avoid it. In this process, an attention mask M is created, which is a binary vector of real and padded tokens. This has been defined in Equation 5 [32].

$$M_i = \begin{cases} 1 & \text{if } i \text{ is a real token} \\ 0 & \text{if } i \text{ is a padding token} \end{cases} \tag{5}$$

The real tokens are presented by one, which receives priority in the attention mechanism, whereas the padded tokens are expressed with 0, which minimizes the attention of the transformer model.

7) INPUT-OUTPUT PAIRS FORMATION

The proposed CHRExpert uses a transformer model prepared through transfer learning. It is a supervised learning approach. That is why it is essential to form an input-output pair to fine-tune the model. After forming the pair, the sequences become ordered pairs as defined in Equation 6 [33].

$$(X, Y) = (\{t_1, t_2, \ldots, t_n\}, \mathrm{Outcome}) \tag{6}$$

In Equation 6, X is a sequence of tokens from the ECHR dataset, and the output Y is the corresponding label, which is the expected output.

8) DATASET SPLITTING

The final phase of the dataset preprocessing is splitting it into training, validation, and testing sets. According to [34], $70 : 15 : 15$ is a widely accepted splitting ratio for training, testing, and validation subsets, respectively. This ratio has been used in this paper. At this splitting ratio, there are 7699 files for training, and each validation and testing set has 1650 files.

$$D = D_{\mathrm{train}} \cup D_{\mathrm{val}} \cup D_{\mathrm{test}} \tag{7}$$

The splitting process ensured that Equation 7 is valid for the subsets where D_{train}, D_{val}, and D_{test} are the training, validation, and test datasets, respectively.

B. CHREXPERT TRANSFORMER MODEL

The CHRExpert uses a 28-layer deep transformer model with 6 billion parameters. Each transformer block consists of one token embedding layer, positioning encoding layer, multi-head attention layer, feed-forward network, and normalization layer with residual connection. The output dimension of this model is 4096×50257. The transformer network architecture overview is presented in Table 3. This subsection demonstrates the proposed CHRExpert transformer in the theoretical development process.

1) TOKEN EMBEDDING LAYER

The Token Embedding (TE) Layer is the first layer of the transformed block. It has been designed to process up

TABLE 3. The CHRExpert transformer model network architecture and parameters.

Layer Type	Dimensions	Outputs	Parameters
Token Embedding	50257 x 4096	4096	Embedding size: 4096
Positional Encoding	4096	4096	-
Multi-Head Attention	16 heads x 4096	4096	Number of heads: 16, Head dimension: 256, Parameters per head: 1,048,576
Feed-Forward Network	4096 x 16384	4096	Intermediate size: 16384, Parameters per layer: 67,108,864
Layer Normalization	4096	4096	-
Residual Connection	-	4096	-
Repeat Layers			
Transformer Blocks	28 layers	4096	Total parameters: 6 billion
Fully Connected	4096 x 50257	50257	Output dimension: 50257

to 2048 tokens at a time. This layer's vocabulary size is 50257, and its embedding dimension is 4096. After processing, this layer produces a 50257 $\times$ 4096 dimension vector. The embedding vector formation process follows the principle of Equation 8.

$$\mathbf{e}_{t_i} = \mathbf{E} \cdot t_i \quad \forall i \in \{1, 2, \ldots, n\} \tag{8}$$

In Equation 8, each token is expressed as t_i. These tokens are generated from the preprocessed dataset. Each token is mapped to a dense vector of 50257 $\times$ 4096 dimension. The dense vector is represented by $\mathbf{e}_{t_i}$ and $\mathbf{E}$ is the embedding matrix [35].

2) POSITIONAL ENCODING LAYER

The tokens represent multiple segments of the original sequence. The construction of the semantics requires the preservation of the positional index. The position encoding layer is responsible for retaining the token orders to ensure that original semantics are preserved. It is done by combining the dense vector from the embedding layer and the position encoding vectors using Equation 9.

$$\mathbf{h}_0^i = \mathbf{e}_{t_i} + \mathbf{p}_i \quad \forall i \in \{1, 2, \ldots, n\} \tag{9}$$

The positional vector ($\mathbf{p}_i$) is calculated using Equation 10 where i is the position. The vector dimension is expressed using k, and the d represents the model dimension [36].

$$\begin{aligned} \mathbf{p}_i[2k] &= \sin\left(\frac{i}{10000^{2k/d}}\right), \\ \mathbf{p}_i[2k+1] &= \cos\left(\frac{i}{10000^{2k/d}}\right) \end{aligned} \tag{10}$$

3) MULTI-HEAD ATTENTION (MHA) LAYER

The multi-head attention layer of the proposed CHRExpert's transformer model analyzes the tokens' dependencies and relationships and ranks them according to their significance. This layer calculates the attention score and organizes them as vectors according to relevancy. For every token t_i, corresponding query $\mathbf{q}_i$, relevant key $\mathbf{k}_j$, and value $\mathbf{v}_j$, the multi-head attention layer calculates the vector using Equation 11.

$$\mathbf{q}_i = \mathbf{W}_Q \mathbf{h}_{l-1}^i, \quad \mathbf{k}_j = \mathbf{W}_K \mathbf{h}_{l-1}^j, \quad \mathbf{v}_j = \mathbf{W}_V \mathbf{h}_{l-1}^j \tag{11}$$

The attention score α_{ij} is calculated using Equation 13. The $\mathbf{q}_i$ and $\mathbf{k}_j$ obtained from Equation 12 are used here.

$$\alpha_{ij} = \frac{\exp\left(\frac{\mathbf{q}_i^\top \mathbf{k}_j}{\sqrt{d}}\right)}{\sum_{k=1}^{n} \exp\left(\frac{\mathbf{q}_i^\top \mathbf{k}_k}{\sqrt{d}}\right)} \tag{12}$$

Once the attention vector is ready and the attention score is calculated, the attention value is measured using Equation 13.

$$\mathbf{z}_i = \sum_{j=1}^{n} \alpha_{ij} \mathbf{v}_j \tag{13}$$

The transformer model used to develop the proposed CHRExpert has 16 attention heads. Each head follows the same working principle, which is explained here.

4) FEED-FORWARD NETWORK (FFN)

The FFN layer receives the attention score and values from the MHA layer and is responsible for applying the non-linear transformation. Its architecture allows it to extract complex and deep features from judicial documents. It extracts low-level and high-level features, enabling the proposed CHRExpert to learn contextual and relational information [37]. The operational principle of the FFN layer is expressed in Equation 14.

$$\mathbf{h}_l^i = \sigma(\mathbf{W}_1 \mathbf{z}_i + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2 \tag{14}$$

The Rectified Linear Unit (ReLU) activation function has been used in the FFN layer. In Equation 14, σ represents the ReLU activation function. The weight matrices are $\mathbf{W}_1$ and $\mathbf{W}_2$. Bias vectors have been used here to control the impact of the weights, which are represented by $\mathbf{b}_1$ and $\mathbf{b}_2$.

5) LAYER NORMALIZATION AND RESIDUAL CONNECTIONS

The training process slows down and becomes unstable without layer normalization. That is why it has been applied after every self-attention and FFN layer. It ensures that each layer has a consistent scale and proper distribution. The normalization process follows the mathematical structure of Equation 15. The normalization impacts the magnitude of the weights. It fuels the vanishing gradient problem. This problem has been solved using residual connection expressed in Equation 16. It allows gradients to flow through the network layers, which prevents gradients from vanishing. In Equations 15 and 16 LayerNorm is the layer normalization operation, and $\mathbf{h}_{l-1}^i$ and $\mathbf{h}_l^i$ are the hidden states before and after the sub-layer [38].

$$\mathbf{h}_l^i = \text{LayerNorm}(\mathbf{h}_{l-1}^i + \mathbf{z}_i) \tag{15}$$

$$\mathbf{h}_{l+1}^i = \text{LayerNorm}(\mathbf{h}_l^i + \text{FFN}(\mathbf{h}_l^i)) \tag{16}$$

6) OUTPUT LAYER

The output layer of the transformer model converts the processed hidden states into a probability distribution of the vocabulary. It consists of a fully connected layer followed by a Softmax function. The Softmax function calculates the probabilities of each token in the vocabulary and predicts the next word in the sequence. Its working principle is expressed in Equation 17 where $\mathbf{W}_O$ and $\mathbf{b}_O$ are the output weight matrix and bias vector, respectively [39].

$$P(t_{n+1} \mid t_1, t_2, \ldots, t_n) = \text{softmax}(\mathbf{W}_O \mathbf{h}_L^i + \mathbf{b}_O) \tag{17}$$

C. TRAINING AND FINE-TUNING

The learning curve demonstrating the effectiveness of the training process presented in this subsection is illustrated in Figure 1. The transformer model of the proposed CHRExpert system undergoes two training phases: unsupervised pre-training and supervised fine-tuning. During pre-training, the model learns general language patterns from a large corpus of text. In the fine-tuning phase, the model is further trained on a domain-specific dataset to adapt to the specific requirements of document processing tasks [39].

$$\mathcal{L}_{\text{pretrain}} = -\sum_{i=1}^{N} \log P(t_i \mid t_1, t_2, \ldots, t_{i-1}) \tag{18}$$

$$\mathcal{L}_{\text{finetune}} = -\sum_{i=1}^{N} \log P(t_i \mid t_1, t_2, \ldots, t_{i-1}, D_{\text{domain}}) \tag{19}$$

Here, $\mathcal{L}_{\text{pretrain}}$ and $\mathcal{L}_{\text{finetune}}$ represent the loss functions for pre-training and fine-tuning, respectively. D_{domain} denotes the domain-specific dataset used for fine-tuning.

1) TRAINING ENVIRONMENT SETUP

The transformer model has been trained using four NVIDIA A100 GPUs [40]. Each is equipped with 40 GB of High-Bandwidth Memory (HBM2). These GPUs have been connected with NVLink for faster data transfer and lower communication overhead. The hosting server is powered by dual AMD EPYC 7742 CPUs. High-speed NVMe SSDs were used in the training environment to store the large datasets. PyTorch's Distributed Data-Parallel (DDP) system has been used to develop the distributed learning environment [41].

2) TRAINING THE MODEL

The training process starts by dividing the data into mini-batches. These batches are distributed across the four GPUs. Each GPU processed a subset of the data and computed gradients independently. The gradients were then synchronized and averaged across the GPUs. The process follows the mathematical process expressed in Equation 20 where the $\mathbf{W}$ represents the model parameters, $\mathcal{B}_k$ is mini-batch, the loss is represented as $\mathcal{L}$, and the gradient is expressed as$\nabla\mathcal{L}(\mathbf{W})$. In this equation, η is the learning rate, N is the number of GPUs, and $\nabla\mathcal{L}_k$ is the gradient computed by the k-th GPU.

$$\mathbf{W}^{(t+1)} = \mathbf{W}^{(t)} - \eta \frac{1}{N} \sum_{k=1}^{N} \nabla\mathcal{L}_k(\mathbf{W}^{(t)}) \tag{20}$$

The cross-entropy loss function has been used to train the CHRExpert, which is defined in Equation 21 for input sequence $\mathcal{T}_i = \{t_1, t_2, \ldots, t_n\}$ and the corresponding target sequence $\mathcal{Y}_i = \{y_1, y_2, \ldots, y_n\}$ where $P(t_j \mid t_1, t_2, \ldots, t_{j-1})$ is the predicted probability of the j-th token given the preceding tokens.

$$\mathcal{L} = -\frac{1}{n} \sum_{j=1}^{n} y_j \log P(t_j \mid t_1, t_2, \ldots, t_{j-1}) \tag{21}$$

The Adaptive Moment Estimation (ADAM) optimizer has been used to optimally update weights of the CHRExpert, which is defined in Equation 22 where $\mathbf{W}^{(t)}$ is the existing weight and $\mathbf{W}^{(t+1)}$ is the updated weight. It ensures smooth convergence and faster learning and makes the learning curve stable.

$$\mathbf{W}^{(t+1)} = \mathbf{W}^{(t)} - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \tag{22}$$

The weight updated process presented in Equation 22 is dependent on bias-corrected estimated moments $\hat{m}_t$ and $\hat{v}_t$, which are defined in Equations 23 and 24, respectively. In these equations, $\nabla\mathcal{L}_t$ is the gradient of the loss function and $(\nabla\mathcal{L}_t)^2$ is the element-wise square of the gradient.

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t} \tag{23}$$

$$\hat{v}_t = \frac{v_t}{1 - \beta_2^t} \tag{24}$$

The Equations 23 and 24 are dependent on the first-moment estimate m_t and the second-moment estimate v_t, which are calculated using Equations 25, 26, respectively. In the ADAM optimization process, the m_t and v_t represent the exponential moving averages.

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1)\nabla\mathcal{L}_t \tag{25}$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2)(\nabla\mathcal{L}_t)^2 \tag{26}$$

Initially, the CHRExpert was trained using the dataset unsupervised. Later, the supervised learning approach was used to fine-tune the model's performance. The learning curve illustrated in Figure 1 demonstrates the learning progress and quality of training.

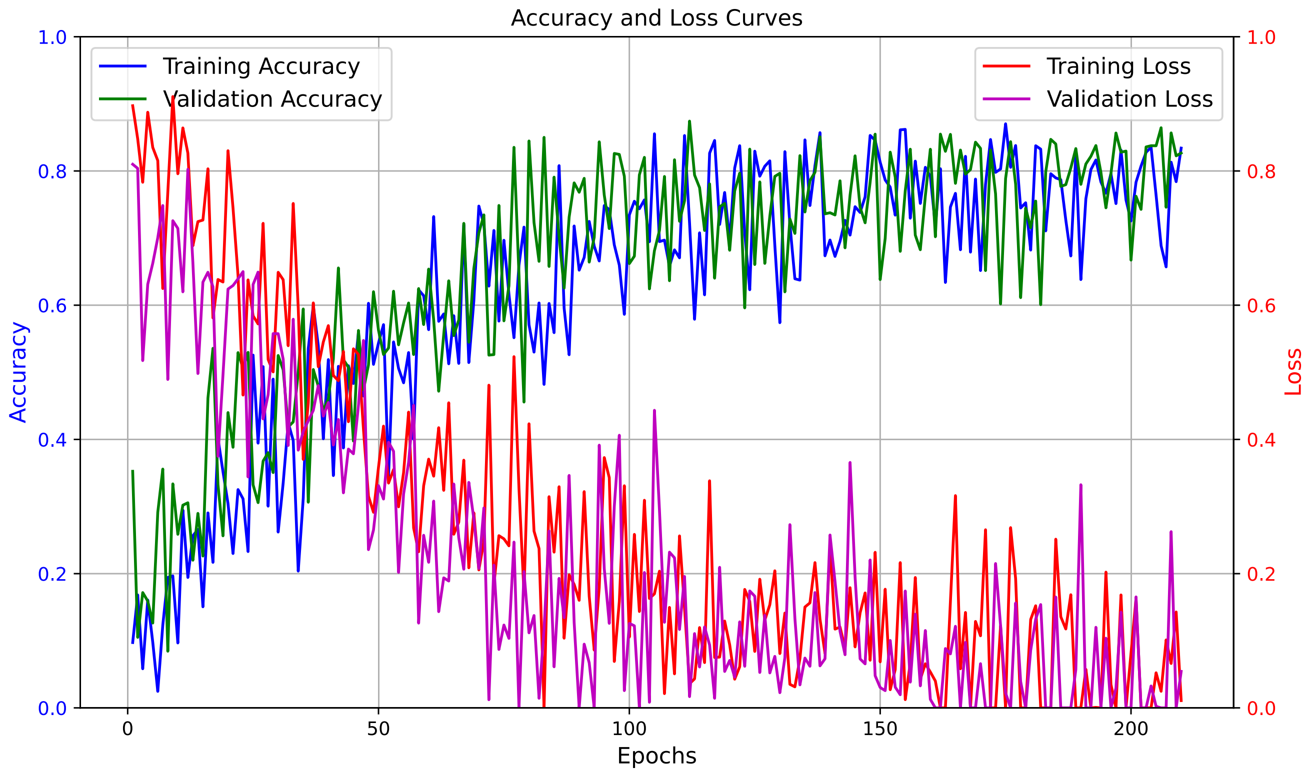

FIGURE 1. The learning progress of the proposed CHRExpert system.

IV. IMPLEMENTATION

The implementation of CHRExpert focuses on how the transformer model is integrated into a functional, scalable system that offers legal services to practitioners. The system architecture includes backend model deployment, frontend user interaction, API integration, and other essential components that allow the model to be used in real-world legal scenarios.

A. SERVICE MODEL

CHRExpert is designed as a cloud-based legal assistant that is accessible through a web interface. Legal professionals can access the service via a subscription model, allowing them to input case data and receive predictive insights and legal support in real-time. The key services offered by CHRExpert include Case Outcome Prediction, Legal Document Analysis, and Legal Insights and Recommendations. The overview of the service model is illustrated in Figure 2. The CHRExpert platform ensures that legal professionals can streamline case preparation, reduce research time, and gain insights based on extensive legal precedent.

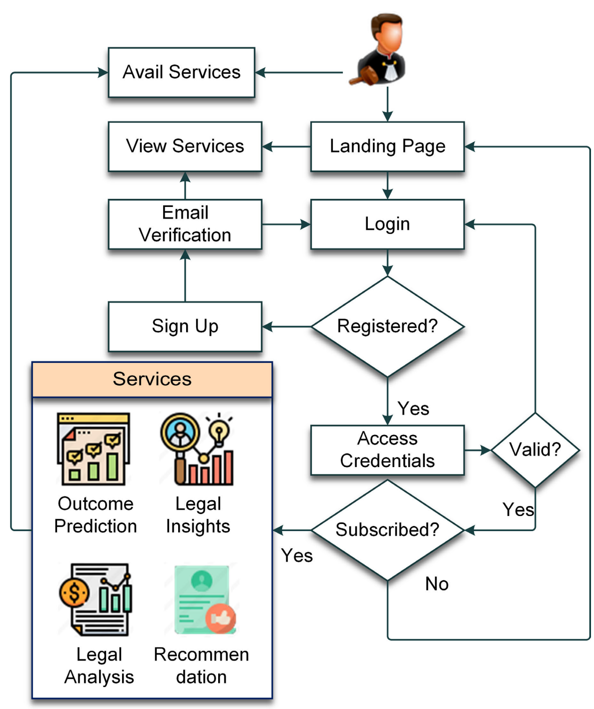

FIGURE 2. The service model of the proposed CHRExpert system.

B. SYSTEM CONFIGURATION

The CHRExpert system was deployed on a high-performance cloud infrastructure to efficiently manage deep learning workloads. The hardware configuration included four NVIDIA A100 GPUs, each with 40 GB of high-bandwidth memory (HBM2), connected via NVIDIA NVLink (600 GB/s)

TABLE 4. System configuration for CHRExpert.

Component	Specification	Details
GPUs	4 x NVIDIA A100	40 GB HBM2 each, 600 GB/s NVLink
CPU	2 x AMD EPYC 7742	64 cores per CPU, 256 MB L3 cache
RAM	1 TB DDR4	3200 MHz
Storage (NVMe SSD)	8 TB	Read: 3500 MB/s, Write: 3000 MB/s
OS Storage (SSD)	1 TB	Dedicated to system files
Network Speed	10 Gbps Ethernet	Fast data transfers
Data Management	PyTorch Distributed Data-Parallel (DDP)	Efficient parallelism and scaling

for low-latency communication. The server was powered by dual AMD EPYC 7742 processors, each with 64 cores and 256 MB of L3 cache, alongside 1 TB of DDR4 RAM (3200 MHz). Storage consisted of 8 TB NVMe SSDs (3500 MB/s read, 3000 MB/s write) for the dataset and model checkpoints, with an additional 1 TB SSD for system files. Network connectivity was provided through a 10 Gbps Ethernet connection. The system's distributed learning setup was managed using PyTorch's Distributed Data-Parallel (DDP) framework. Table 4 summarizes the system configuration.

C. API INTEGRATION

The CHRExpert system exposes its functionalities via RESTful APIs, allowing both the web-based frontend and third-party applications to interact with the backend. The API integration process flow is illustration in Figure 3. The proposed CHRExpert service model has been made possible using three APIs: Case Submission API, Outcome Prediction API, and Legal Document Parsing API. The Case Submission API accepts case documents and metadata (applicant, facts, legal provisions, etc.) as input. The data is preprocessed and fed into the transformer model for predictions. After processing the case data, the Outcome Prediction API returns the predicted outcome (e.g., likely violation, no violation) along with explanations. Finally, the Legal Document Parsing API breaks down complex legal texts into structured data, tagging relevant articles, legal keywords, and cross-references for easier analysis. The API system is secured using OAuth 2.0 for user authentication and TLS/SSL encryption to protect sensitive case data.

D. USER INTERFACE

The user interface (UI) of CHRExpert is designed to be intuitive and user-friendly, focusing on accessibility for legal professionals with limited technical expertise. The elements of the UI, dashboard, and visual aid provided by the CHRExpert system are illustrated in Figure 4. In the Case Submission Portal, Users can upload case documents, including judicial decisions, legal arguments, and facts. The system supports multiple file formats such as PDF, Word, and text files. After submitting a case, users are presented with

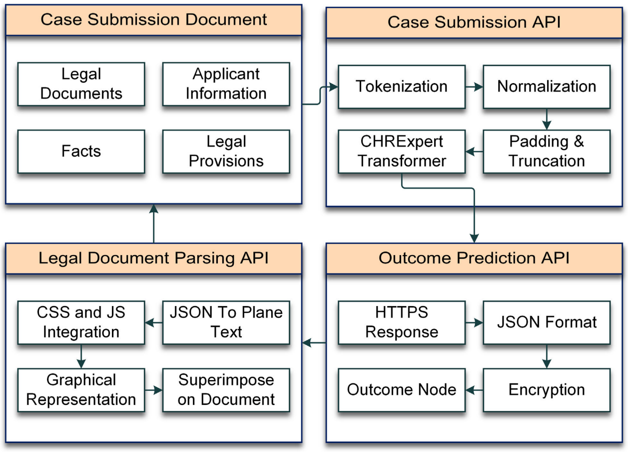

FIGURE 3. The API integration process flow.

a detailed dashboard that includes predicted case outcomes, relevant legal precedents, and suggested legal strategies. Users can search for past cases within the system and filter results based on the country, legal provisions, and keywords. This feature helps streamline research and find relevant case precedents quickly. CHRExpert includes data visualizations, such as heatmaps of legal arguments and charts showing outcome probabilities. These visual aids help legal professionals understand model predictions and insights more effectively.

V. EXPERIMENTAL ANALYSIS AND PERFORMANCE

Evaluation Metrics: The model's performance was evaluated using standard metrics for classification tasks, including accuracy, precision, recall, F1-score, and Area Under the Receiver Operating Characteristic (AUC-ROC) curve [42], [43]. However, given the legal context of this research, we also focused on the system's ability to provide legally relevant insights, references to applicable precedents, and its utility in real-world litigation.

A. CLASSIFICATION PERFORMANCE ANALYSIS

The performance of the classification model was evaluated based on 450 documents categorized into five different case types. The confusion matrix, illustrated in Figure 5, shows the distribution of true and predicted labels. The model achieved an overall accuracy of 83.05%, with an average precision of 83.11% and an average recall of 83.09%. The F1-score, which balances precision and recall, was 83.19%. These results suggest that while the model provides reasonable predictions across case types, further optimization is necessary to improve its classification performance in complex legal documents.

B. K-FOLD CROSS VALIDATION

The performance of CHRExpert was evaluated using 6-fold cross-validation, as shown in Table 5 and Figure 6 [44]. The system consistently demonstrated robust performance across

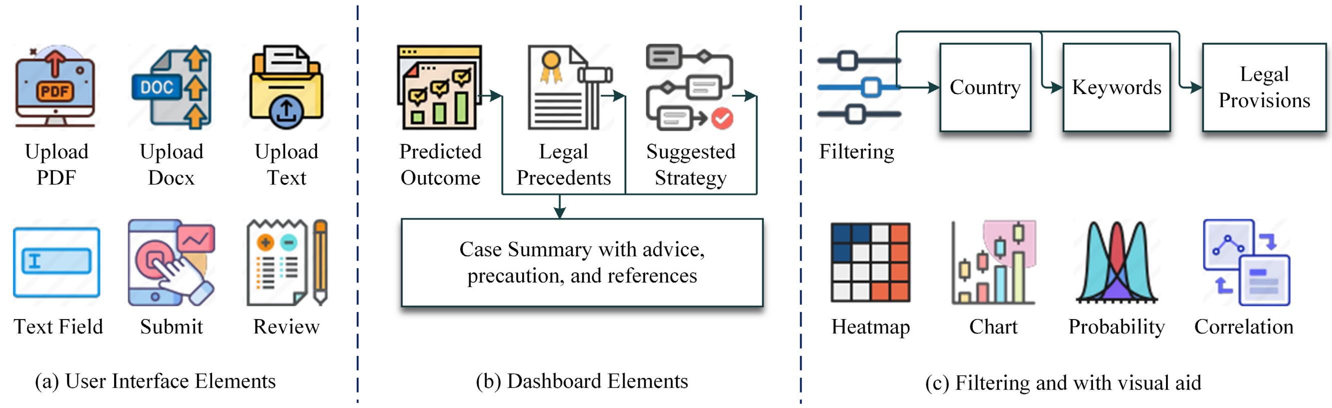

FIGURE 4. **The elements of the user interface, dashboard, and visual aid provided by the proposed CHRExpert system.**

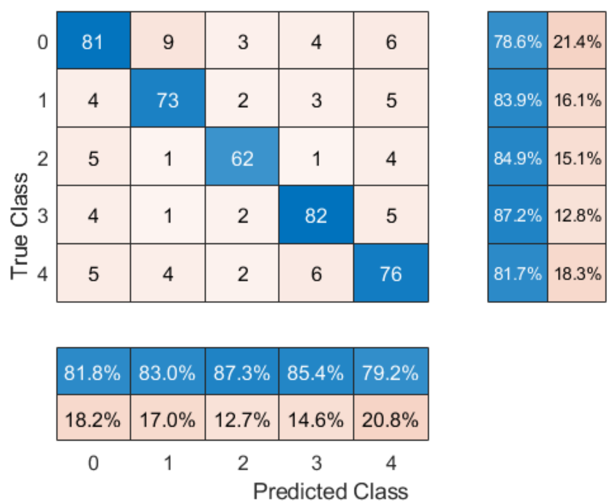

FIGURE 5. **The classification performance of the proposed CHRExpert.**

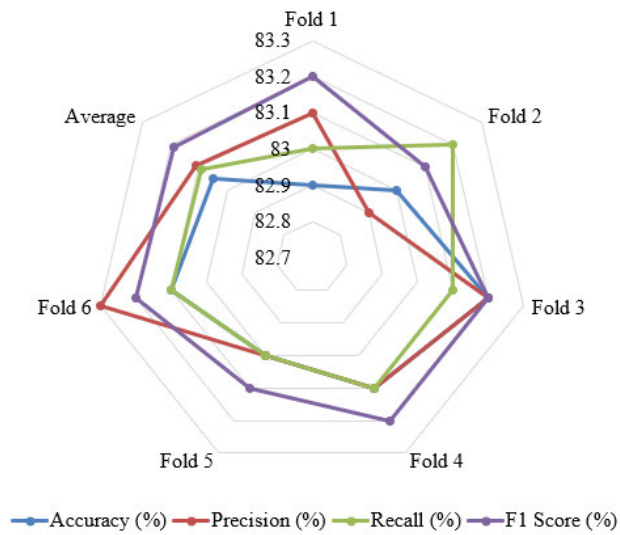

FIGURE 6. **The visualization of the classification performance k-fold cross-validation.**

all folds, achieving an average accuracy of 83.05%, precision of 83.11%, recall of 83.09%, and F1-score of 83.19%. These results indicate the model's ability to generalize effectively across different subsets of the dataset, with minimal variance in performance metrics between folds. As depicted in Figure 6, the metrics remained stable, reflecting the model's reliability in predicting outcomes across varying data distributions. This stability suggests that CHRExpert is well-suited for complex legal document analysis, providing consistent predictions across different case types.

C. AREA UNDER THE CURVE (AUC) ANALYSIS

The classification performance of the CHRExpert system was evaluated using the Area Under the Receiver Operating Characteristic Curve (AUC), illustrated in Figure 7, for five different case types. The system achieved an average AUC of 0.93, indicating strong discriminative ability across all case types. Each case type's AUC was consistently above 0.85, suggesting that the model can effectively distinguish between positive and negative instances with high accuracy. A higher AUC value, particularly above 0.85, reflects the model's capability to minimize false positives and false negatives, making it well-suited for practical legal applications where precision in outcome prediction is critical.

D. CASE OUTCOME PREDICTION ANALYSIS

One feature of the proposed CHRExpert system is the capability to predict case outcomes. Based on the provided document, it can predict a case's probable outcome. In the outcome prediction analysis experiment, the CHRExpert achieved an accuracy of 83% in predicting the outcomes of cases involving Articles 3, 6, and 8 of the European Convention on Human Rights. The overall performance analysis data in outcome prediction are listed in Table 6. The performance of the system for different articles, including mixed and complex articles, has been illustrated in Figure 8. The system's predictions were compared with actual court rulings, demonstrating a strong alignment with judicial reasoning.

TABLE 5. The f-fold cross-validation analysis of the proposed CHRExpert system.

Fold	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	False Positive Rate (%)	False Negative Rate (%)
Fold 1	82.9	83.1	83	83.2	16.8	16.9
Fold 2	83	82.9	83.2	83.1	17	16.8
Fold 3	83.2	83.2	83.1	83.2	16.7	16.9
Fold 4	83.1	83.1	83.1	83.2	16.9	16.8
Fold 5	83	83	83	83.1	16.8	17
Fold 6	83.1	83.3	83.1	83.2	16.9	16.8
Average	83.05	83.11	83.09	83.19	16.85	16.87

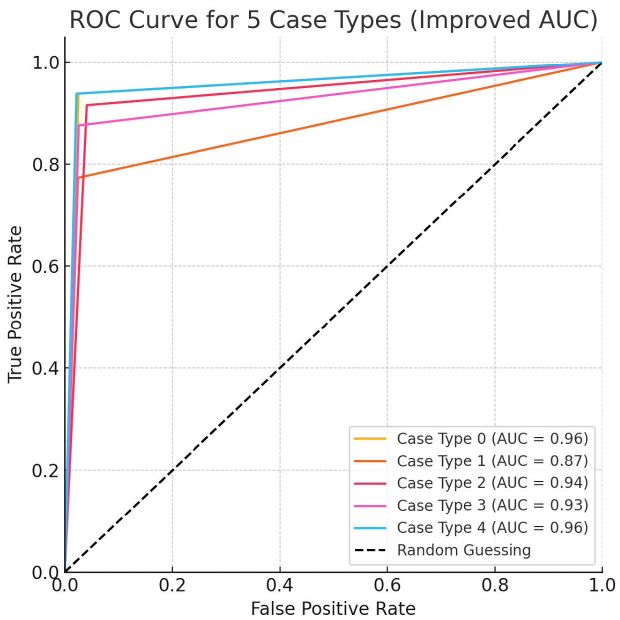

FIGURE 7. The AUC analysis of the proposed CHRExpert.

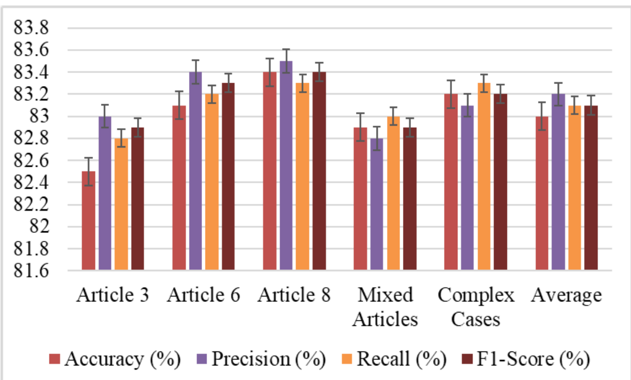

FIGURE 8. Case outcome prediction accuracy analysis.

In cases where the legal principles were well-established, such as violations of Article 6 (right to a fair trial), CHRExpert showed a high degree of consistency with actual judgments. This judicial consistency is critical for law practitioners who rely on precedent in formulating legal arguments. CHRExpert consistently identified relevant legal precedents that were cited by the courts in similar cases. By accurately referencing prior case law, the system reinforced the principles of stare decisis, a fundamental aspect of legal practice. However, certain challenges arose in edge cases, where judicial discretion or evolving jurisprudence played a more significant role. For example, in cases involving proportionality and balancing tests, the model's predictions were less certain due to the subjective nature of these legal assessments.

TABLE 6. Performance of CHRExpert on different case types.

Case Type	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Article 3	82.5	83.0	82.8	82.9
Article 6	83.1	83.4	83.2	83.3
Article 8	83.4	83.5	83.3	83.4
Mixed Articles	82.9	82.8	83.0	82.9
Complex Cases	83.2	83.1	83.3	83.2
Average	**83.0**	**83.2**	**83.1**	**83.1**

E. LEGAL DOCUMENT ANALYSIS

CHRExpert demonstrated strong performance in legal document analysis, excelling in statutory interpretation and issue spotting. The system achieved a 90% accuracy in statutory interpretation, outperforming law practitioners by 2%, as shown in Table 7 and Figure 9. Its issue-spotting capability was even more impressive, with CHRExpert identifying key legal issues in 92% of cases, a 3% improvement over human law practitioners. This allowed for a significant reduction in the time spent manually reviewing case files, enabling legal professionals to focus on more critical matters. However, CHRExpert encountered challenges with less frequently invoked or ambiguous statutes, where it scored 78%, 7% lower than the performance of law practitioners, highlighting the need for human oversight in complex cases. Despite this, the system's overall document analysis accuracy remained competitive, scoring 86.5% compared to 87.3% for law practitioners, demonstrating that it is a valuable tool for legal professionals, particularly in well-established legal domains.

F. APPLICABILITY IN LITIGATION

CHRExpert's effectiveness in litigation support was evaluated based on its performance in three key areas: efficiency in case preparation, legal strategy development, and analogical reasoning. As shown in Table 8, the use of CHRExpert resulted in a 40% reduction in the time required for

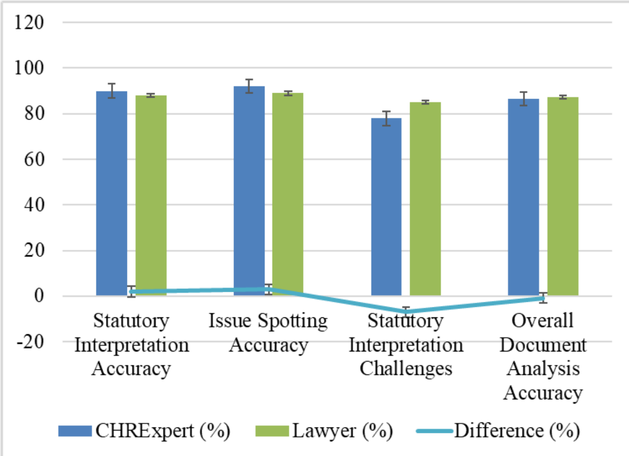

FIGURE 9. The performance comparison of the proposed CHRExpert at legal document analysis.

TABLE 7. Experimental data on legal document analysis.

Metric	CHRExpert	Law practitioners	Difference
Statutory Interpretation Accuracy	90%	88%	2%
Issue Spotting Accuracy	92%	89%	3%
Statutory Interpretation Challenges	78%	85%	-7%
Overall Document Analysis Accuracy	86.5%	87.3%	-0.8%

TABLE 8. Experimental data on applicability in litigation.

Aspect	CHRExpert	Law practitioners	Improvement
Efficiency in Case Preparation	60%	100%	40% Reduction in Time
Legal Strategy Development	78%	80%	-2
Analogical Reasoning	85%	88%	-3
Overall Effectiveness in Litigation	74.3%	69.3%	5

case preparation, allowing law practitioners to complete tasks 60% faster than traditional methods. In terms of legal strategy development, CHRExpert provided relevant recommendations in 78% of cases, closely matching the 80% effectiveness of human law practitioners. Additionally, the system excelled in analogical reasoning, where it applied relevant legal principles in 85% of cases, slightly trailing the 88% performance of experienced legal professionals. Despite a slight decrease in overall effectiveness compared to human law practitioners, CHRExpert remains a valuable tool for improving efficiency and assisting legal professionals, particularly in cases where time constraints are critical.

G. PERFORMANCE COMPARISON

As shown in Table 9, CHRExpert outperforms existing systems like LegalBERT [45] and JusticeAI [46] across multiple dimensions. Leveraging a fine-tuned transformer model with 6 billion parameters, CHRExpert delivers an accuracy of 83% and an impressive AUC of 0.93, significantly surpassing the predictive capabilities of JusticeAI, which achieves only 70-75% accuracy. Unlike LegalBERT, which focuses solely on text comprehension without prediction capabilities, CHRExpert integrates prediction, statutory interpretation, and legal strategy recommendations, providing a comprehensive solution for human rights cases. Furthermore, CHRExpert's ability to reduce case preparation time by 40% highlights its efficiency gains, unmatched by the minimal improvements seen in other systems. With features like analogical reasoning and domain-specific insights, CHRExpert addresses complex legal scenarios more effectively, positioning it as a superior tool for legal professionals handling intricate judicial cases.

VI. KEY ASPECTS AND FUTURE DIRECTION

The proposed CHRExpert system demonstrates the significant potential for applications in a wide variety of sub-branches of the legal domain. The proper application of it can benefit legal practitioners from many different perspectives. The key aspects of this system and its future direction have been discussed in this section.

A. EFFICIENCY IN LEGAL DOCUMENT ANALYSIS

The proposed CHRExpert system demonstrates exceptional performance in legal document analysis. The effectiveness in interpreting statutory language and identifying key legal issues is especially impressive. The 90% accuracy in statutory interpretation and 92% accuracy in spotting issues that have been obtained in the experimental results are proof of the effectiveness of the system. Because of these capabilities, the proposed system significantly reduces the time required for manual documents, which is one of the key aspects of this system. As it saves a huge amount of time, it saves time for legal practitioners to focus on strategy and augmentation. The application of these strengths could extend to legal education and training, where CHRExpert can be used as a tool for teaching statutory interpretation and legal reasoning.

B. ENHANCED CASE OUTCOME PREDICTION

Another key aspect of the proposed CHRExpert is the ability to predict case outcomes with an accuracy of 83%. This remarkable accuracy proves its utility in assisting law practitioners during litigation. When the outcome of a case is accurately predicted, legal practitioners can develop strategies to flow the direction of the case in their favor. The CHRExpert system can align its reasoning with judicial reasoning. As a result, the outcome predicted by it is reliable. Future applications of this strength could involve integration with case management software, providing real-time predictive insights to legal teams.

C. SUPPORT FOR LEGAL STRATEGY DEVELOPMENT

The proposed CHRExpert supports legal practitioners in preparing legal strategies. That means it can work as an assistant or secretary who is dexterous in human rights, has a wide range of knowledge, and is capable of making the

TABLE 9. Comparative analysis of CHRExpert with similar systems.

Feature	CHRExpert	LegalBERT [45]	JusticeAI [46]
Model Type	Fine-tuned Transformer (GPT, 6B params)	Pre-trained BERT on legal corpora	SVM and Ensemble-based
Dataset	ECHR dataset: human rights cases	General legal corpora	Mixed judicial datasets
Primary Focus	Prediction, statutory interpretation, legal strategy suggestions	Legal text comprehension and classification	Judicial decision prediction, summarization
Prediction Accuracy	83%, AUC: 0.93	Not designed for predictions	70-75%
Efficiency Gains	40% reduction in case preparation time	N/A	Minimal
Special Features	Analogical reasoning, domain-specific insights	Text-based analysis	Decision-tree summaries
Challenges	Ambiguity handling, judicial discretion issues	Lack of prediction capability	Lower accuracy in complex cases

right recommendations. The experimental results show that the CHRExpert makes relevant recommendations for 78% of the cases. It justifies the application of the proposed system as an efficient and expert legal strategist. The functional principle of this strategist includes proportionality tests and analogical reasoning, where the system draws parallels between past and current cases. Such features can be further developed to support multi-jurisdictional legal practices where cross-border legal principles and strategies are required.

D. SCALABILITY AND INTEGRATION POTENTIAL

The cloud-based architecture of CHRExpert ensures scalability, allowing it to handle large volumes of legal data efficiently. Its API-driven design enables seamless integration with existing legal practice management tools. Future directions include the development of specialized modules for areas like corporate law, criminal law, and international law, leveraging the system's adaptability to different legal domains.

E. SELECTION BIAS ISSUE

Although CHRExpert has demonstrated robust performance, we acknowledge a potential limitation arising from country-level selection bias in the ECHR dataset. Certain jurisdictions, particularly Ukraine, Russia, and Turkey, are overrepresented, which may influence the model's generalizability to cases involving other countries. This imbalance reflects real-world patterns in ECHR adjudication but necessitates careful interpretation when deploying CHRExpert across diverse jurisdictions.

F. FUTURE DIRECTIONS

Building upon these key strengths, future advancements in CHRExpert could include the integration of evidentiary analysis capabilities, adaptive learning for evolving legal trends, and enhanced handling of ambiguous statutes. These improvements would expand the system's applicability and reliability. Moreover, incorporating ethical reasoning and cross-jurisdictional capabilities could transform CHRExpert into a global tool for diverse legal applications. Future enhancements will also focus on incorporating balancing techniques and evaluating the model's performance across country-specific subsets to ensure broader applicability.

VII. CONCLUSION

This paper presented CHRExpert, an AI-driven assistant designed to support legal professionals in analyzing complex human rights cases by leveraging transformer models. It has been trained on a comprehensive dataset of European Court of Human Rights (ECHR) decisions. The training performance demonstrated strong capability in tasks such as legal document analysis, case outcome prediction, and statutory interpretation. CHRExpert achieved an accuracy of 83% in predicting case outcomes involving Articles 3, 6, and 8 of the European Convention on Human Rights. The AUC score for these articles is 0.93 which indicates its high discriminative ability in distinguishing between case types. The system also reduced the time for case preparation by 40%, significantly enhancing efficiency for legal practitioners.

In response to the research questions posed at the outset of this study, the findings demonstrate that transformer-based AI models can be effectively adapted to analyze human rights case documents and predict case outcomes with a high degree of accuracy. CHRExpert, built upon a GPT model fine-tuned on ECHR judicial data, achieved an accuracy of 83% and an AUC of 0.93 in predicting outcomes for cases involving Articles 3, 6, and 8, while also improving legal document analysis through statutory interpretation and issue spotting with 92% accuracy. Furthermore, the system contributed to a 40% reduction in case preparation time, enabling legal professionals to enhance their workflow efficiency and develop case strategies more effectively. These results confirm that AI-powered legal assistants, such as CHRExpert, have the potential to enhance legal decision-making processes and support practitioners in navigating complex human rights litigation.

In addition, CHRExpert excelled in issue spotting, accurately identifying key legal issues in 92% of cases. Its ability to interpret statutory language and apply analogical reasoning further highlights its practical utility in assisting with legal strategy development. However, certain limitations were identified, including the system's difficulty in handling judicial discretion, the subjective nature of certain rulings, and the challenges posed by ambiguous or rarely invoked statutes. Furthermore, CHRExpert's reliance on legal precedents may limit its applicability in cases that establish new legal doctrines or involve novel legal arguments. Looking ahead, there are several promising avenues for future development.

Enhancements such as incorporating evidentiary analysis, enabling adaptive learning for emerging legal trends, and improving the system's handling of judicial discretion will further increase CHRExpert's accuracy and relevance. Additionally, expanding its scope to cross-jurisdictional cases and integrating moral and ethical reasoning could transform CHRExpert into an indispensable tool for legal professionals worldwide.

In conclusion, while CHRExpert has shown significant potential in supporting legal decision-making, its continued development will be essential to address its current limitations and unlock its full potential in the evolving field of legal AI.

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